21 Oct 2026 00:55:20 - cooling_tower_water_makeup_psychrometrics.sm
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Industrial Cooling Tower Psychrometrics, Merkel Heat/Mass Transfer & Water Makeup Model (Osmaniye, Turkey)

2026-10-20

Industrial Induced-Draft Counterflow Cooling Tower Simulation

Heat & Mass Transfer, Exit Air Psychrometrics & Makeup Water Optimization (Osmaniye, Turkey)

Dual Plugins: IF97Wrapper.dll (IAPWS-IF97 Steam Tables) + PsychroLibWrapper.dll (ASHRAE 2017 Psychrometrics)

Strict Physical Unit Guarantee: Pressures [Pa, bar], Temperatures [K, °C], Enthalpies [J/kg, kJ/kg], Flows [kg/s, m³/h]

This worksheet provides a rigorous thermohydraulic, psychrometric, and water-mass balance model for an industrial mechanical-draft counterflow cooling tower operating in Osmaniye, Turkey (elevation z = 125 m).

The cooling tower rejects 10.45 MW of process heat from a chemical plant / power cycle loop, cooling 250 kg/s (900 m³/h) of circulating water from 42.0 °C to 32.0 °C across an ambient summer wet-bulb temperature of 25.83 °C.

Key Engineering Highlights:

1. Dual Plugin Integration: High-accuracy thermodynamic properties of water and steam from IF97 (heat capacity cp, latent heat h_fg, saturation pressure Psat, liquid density) coupled with ASHRAE 2017 moist air psychrometrics (barometric pressure, humidity ratio, wet-bulb, enthalpy).
2. Non-Linear Exit Air Solving: Formulation of tower energy balance and solution of saturated exit air temperature (T_out = 34.33 °C) and humidity ratio (W_out = 35.73 g/kg) via plugin-free dimensional Secant solver.
3. Merkel Transfer Number (Me = KaV/L): Calculation of tower characteristic via Cooling Technology Institute (CTI) 4-point Chebyshev numerical quadrature (Me_cheb = Me_int = 1.0107).
4. Dimensional Native RK4 Fill Depth Profiles: 13-statement Runge-Kutta 4th-order algorithm (Urroz paradigm) formulated inside an AreaRegion with 100% authentic physical units, yielding axial temperature and enthalpy profiles.
5. Comprehensive Water Makeup & Conservation: Parametric evaluation of evaporation loss, drift, blowdown, and total makeup water consumption across Cycles of Concentration (CoC = 2 to 8).

Section 1: Site Altimetry & Osmaniye Ambient Psychrometrics

Barometric reduction to site elevation (z = 125 m) and evaluation of summer design ambient moist air properties.

`Plant site elevation above mean sea level in Osmaniye, Turkey: z = 125 m`

$z_site := 125\ m$

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Site barometric atmospheric pressure: P_atm = 99 832 Pa (0.9983 bar)

$P_atm := \text{Psy_AtmPressure}(z_site) = 99832.3333\ \text{Pa}$

Osmaniye summer design dry-bulb temperature: T_db = 36.0 °C (309.15 K)

$T_db_amb := 36 + 273.15\ \text{K}$

Summer design ambient relative humidity: RH = 45% (0.450)

$RH_amb := 0.45$

Ambient wet-bulb temperature: T_wb = 25.83 °C (298.98 K)

$T_wb_amb := \text{Psy_TWetBulb_RH}(T_db_amb, RH_amb, P_atm) = 298.9778\ \text{K}$

Ambient humidity ratio: W_amb = 0.01713 kg/kg dry air (17.13 g/kg)

$W_amb := \text{Psy_HumRatio_RH}(T_db_amb, RH_amb, P_atm) = 0.0171$

Inlet ambient moist air specific enthalpy: h_in = 80 206 J/kg dry air (80.21 kJ/kg)

$h_a_in := \text{Psy_Enthalpy}(T_db_amb, W_amb) = 80205.9398 \cdot \frac{1}{\text{kg}}\ \text{J}$

Inlet moist air density: rho_in = 1.114 kg/m³

$rho_a_in := \text{Psy_AirDensity}(T_db_amb, W_amb, P_atm) = 1.1136\ \frac{\text{kg}}{\text{m}^3}$

Ambient dew-point temperature: T_dp = 22.19 °C (295.34 K)

$T_dp_amb := \text{Psy_TDewPoint_RH}(T_db_amb, RH_amb) = 295.3426\ \text{K}$

Section 2: Cooling Tower Circulation, Range, Approach & Thermal Duty

Water mass flow, cooling range, thermal approach, and duty evaluation using IAPWS-IF97 liquid properties.

Circulating cooling water mass flow rate: m_w = 250 kg/s (900 m³/h)

$m_w := \frac{250\ \text{kg}}{\text{s}}$

Hot water return temperature from process exchangers to tower top: T_w_in = 42.0 °C (315.15 K)

$T_w_in := 42 + 273.15\ \text{K}$

Cold water supply temperature from basin to plant exchangers: T_w_out = 32.0 °C (305.15 K)

$T_w_out := 32 + 273.15\ \text{K}$

Mean circulating water temperature: T_w_mean = 37.0 °C (310.15 K)

$T_w_mean := 0.5 \cdot T_w_in + T_w_out$

Cooling range: Delta_T_range = 10.0 K (10.0 °C)

$Delta_T_range := T_w_in - T_w_out$

Cooling approach to ambient wet-bulb: Delta_T_app = 6.17 K (6.17 °C approach)

$Delta_T_app := T_w_out - T_wb_amb = 6.1722\ \text{K}$

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Isobaric heat capacity of liquid water from IF97 at 37 °C: cp_w = 4178.7 J/(kg*K)

$cp_w := \text{IF97_cp_Tp}(T_w_mean, P_atm) = 4178.7164 \cdot \frac{1}{K\ kg}\ J$

Latent heat of vaporization from IF97 at 37 °C: h_fg = 2 413 170 J/kg (2413.2 kJ/kg)

$h_fg_37C := \text{IF97_hfg_T}(T_w_mean) = 2.4132 \cdot 10^{6} \cdot \frac{1}{kg}\ J$

Liquid water density from IF97 at 37 °C: rho_w = 993.3 kg/m³

$rho_w := \text{IF97_rho_Tp}(T_w_mean, P_atm) = 993.3354\ \frac{kg}{m^3}$

Total thermal cooling duty: Q_duty = 10 446 791 W (10.447 MW)

$Q_duty := m_w \cdot cp_w \cdot Delta_T_range = 1.0447 \cdot 10^{7}\ W$

Section 3: Air Mass Flow Rate, Exit Enthalpy & Saturated Exit Air State

Tower overall energy balance, exit moist air enthalpy, and non-linear exit temperature solving via FindRoot().

Liquid-to-Gas mass flow ratio (L/G = m_w / m_a): LG_ratio = 1.10

$LG_ratio := 1.10$

Dry air mass flow rate: m_a = 227.27 kg/s dry air (818.18 t/h)

$m_a := \frac{m_w}{LG_ratio} = 227.2727\ \frac{kg}{s}$

Exit moist air specific enthalpy from overall energy balance: h_out = 126 172 J/kg (126.17 kJ/kg)

$h_a_out := h_a_in + \frac{Q_duty}{m_a} = 1.2617 \cdot 10^{5} \cdot \frac{1}{kg}\ J$

Exit air enthalpy balance error: f_exit(T) = h_sat(T) - h_a_out

$$f_exit(T) := \begin{vmatrix} a := \text{Psy_HumRatio_RH}(T, 1, P_atm) \\ b := \text{Psy_Enthalpy}(T, a) \\ b - h_a_out \end{vmatrix}$$

Verification check of f_exit near solution: f_exit(307.5 K) = +138.5 J/kg (> 0 proves sign change)

$f_exit_chk := f_exit(307.5\ K)$

Plugin-Free Dimensional Secant Solver for Saturated Exit Air Temperature:

Iterative root finding on f_exit(T) = 0 without external solver plugins, converging in ~5 iterations.

Secant initial lower bound estimate: T0 = 305.0 K (31.85 °C)

$T0_sec := 305.0\ K$

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Secant initial upper bound estimate: T1 = 310.0 K (36.85 °C)

$T1_sec := 310.0\ \mathrm{K}$

Enthalpy residual at lower estimate T0: fa_sec < 0

$fa_sec := f_exit(T0_sec)$

Enthalpy residual at upper estimate T1: fb_sec > 0

$fb_sec := f_exit(T1_sec)$

Secant iteration counter register

$k_sec := 0$

Convergence tolerance on enthalpy residual: eps = 0.1 J/kg

$eps_sec := \frac{0.1\ \mathrm{J}}{\mathrm{kg}}$

Dimensional Secant solver while-loop converging exit air temperature T1_sec to |fb| < 0.1 J/kg

```
while k_sec < 20
    T2_sec := eval(T1_sec − (fb_sec · T1_sec − T0_sec) / (fb_sec − fa_sec))
    T0_sec := eval(T1_sec)
    T1_sec := eval(T2_sec)
    fa_sec := eval(fb_sec)
    fb_sec := eval(f_exit(T1_sec))
    k_sec := eval(k_sec + 1)
    if |fb_sec| < eps_sec
      break
    else
      0
```

Exit dry-bulb air temperature: T_out = 307.48 K (34.33 °C)

$T_db_out := T1_sec = 307.482\ \mathrm{K}$

Saturated exit humidity ratio: W_out = 0.03573 kg/kg dry air (35.73 g/kg)

$W_out := \mathrm{Psy_HumRatio_RH}(T_db_out, 1, P_atm) = 0.0357$

Exit moist air density at 34.33 °C: rho_a_out = 1.108 kg/m³

$rho_a_out := \mathrm{Psy_AirDensity}(T_db_out, W_out, P_atm) = 1.1079\ \frac{\mathrm{kg}}{\mathrm{m}^3}$

Exhaust induced-draft fan volumetric capacity: V_fan = 205.1 m³/s (738 500 m³/h)

$V_fan := \frac{m_a}{rho_a_out} = 205.1413\ \frac{\mathrm{m}^3}{\mathrm{s}}$

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Section 4: Water Mass Balance, Evaporation, Drift, Blowdown & Makeup Water

Evaporation rate from moisture pick-up, drift loss, blowdown for Cycles of Concentration (CoC = 4), and makeup.

Evaporation mass loss rate: m_evap = 4.226 kg/s (15.21 m³/h)

$m_evap := m_a \cdot (W_out - W_amb) = 4.2264 \ \frac{kg}{s}$

Evaporative water loss in volumetric flow rate: V_evap = 15.21 m³/h

$$V_evap := \frac{m_evap \ \frac{3600 \ s}{hr}}{\frac{1000 \ kg}{m^3}}$$

Evaporation rate cross-check via IF97 latent heat: m_evap_IF97 = 4.329 kg/s (15.58 m³/h)

$m_evap_IF97 := \frac{Q_duty}{h_fg_37C} = 4.3291 \ \frac{kg}{s}$

Drift eliminator efficiency rating (0.005% of circulating water): f_drift = 5e-5

$f_drift := 0.00005$

Drift droplet carry-over loss rate: m_drift = 0.0125 kg/s (0.045 m³/h)

$m_drift := f_drift \cdot m_w = 0.0125 \ \frac{kg}{s}$

Nominal design Cycles of Concentration for Osmaniye makeup water: CoC = 4.0

$CoC_nom := 4.0$

Continuous blowdown discharge rate to maintain dissolved solids balance: m_bd = 1.409 kg/s (5.07 m³/h)

$m_blowdown := \frac{m_evap}{CoC_nom - 1} = 1.4088 \ \frac{kg}{s}$

Total fresh makeup water demand: m_makeup = 5.648 kg/s (20.33 m³/h)

$m_makeup := m_evap + m_blowdown + m_drift = 5.6478 \ \frac{kg}{s}$

Hourly makeup water requirement: V_makeup = 20.33 m³/h

$$V_makeup_hr := \frac{m_makeup \ \frac{3600 \ s}{hr}}{\frac{1000 \ kg}{m^3}}$$

Daily total makeup water consumption: V_makeup_day = 488.0 m³/day

$V_makeup_day := V_makeup_hr \ \frac{24 \ hr}{day}$

Section 5: Merkel Theory & Cooling Tower Transfer Number (Me = KaV/L)

Evaluation of the Merkel cooling tower characteristic integral and CTI 4-point Chebyshev numerical quadrature.

Air operating line enthalpy: h_a(Tw) = h_in + (L/G)*cp_w*(Tw - T_out)

$h_air_op(Tw) := h_a_in + LG_ratio \cdot cp_w \cdot Tw - T_w_out$

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Saturated air enthalpy at water temperature: h_sat(Tw)

$h_sat_w(Tw) := \text{Psy_Enthalpy}(Tw, \text{Psy_HumRatio_RH}(Tw, 1, P_atm))$

Merkel characteristic integrand: cp_w / (h_sat(Tw) - h_a(Tw)) [dimensionless/K]

$$merkel_integrand(Tw) := \frac{cp_w}{h_sat_w(Tw) - h_air_op(Tw)}$$

CTI Chebyshev node 1: Tw1 = 33.0 °C (306.15 K)

$Tw1 := T_w_out + 0.1 \cdot Delta_T_range = 306.15\ K$

CTI Chebyshev node 2: Tw2 = 36.0 °C (309.15 K)

$Tw2 := T_w_out + 0.4 \cdot Delta_T_range = 309.15\ K$

CTI Chebyshev node 3: Tw3 = 38.0 °C (311.15 K)

$Tw3 := T_w_in - 0.4 \cdot Delta_T_range = 311.15\ K$

CTI Chebyshev node 4: Tw4 = 41.0 °C (314.15 K)

$Tw4 := T_w_in - 0.1 \cdot Delta_T_range = 314.15\ K$

Driving enthalpy difference at node 1: dh1 = 33 028 J/kg

$$dh1 := h_sat_w(Tw1) - h_air_op(Tw1) = 33028.3486 \cdot \frac{1}{kg}\ J$$

Driving enthalpy difference at node 2: dh2 = 38 785 J/kg

$$dh2 := h_sat_w(Tw2) - h_air_op(Tw2) = 38784.5911 \cdot \frac{1}{kg}\ J$$

Driving enthalpy difference at node 3: dh3 = 44 248 J/kg

$$dh3 := h_sat_w(Tw3) - h_air_op(Tw3) = 44246.9059 \cdot \frac{1}{kg}\ J$$

Driving enthalpy difference at node 4: dh4 = 55 265 J/kg

$$dh4 := h_sat_w(Tw4) - h_air_op(Tw4) = 55263.475 \cdot \frac{1}{kg}\ J$$

CTI 4-point Chebyshev quadrature transfer number: Me_cheb = 1.0107 (CTI ATC-105 standard benchmark)

$$Me_cheb := \frac{cp_w \cdot Delta_T_range}{4} \cdot \left(\frac{1}{dh1} + \frac{1}{dh2} + \frac{1}{dh3} + \frac{1}{dh4} \right) = 1.0108$$

Merkel Cooling Tower Number: Me_int = 1.0107 (evaluated via CTI standard Chebyshev quadrature)

$Me_int := \text{eval}(Me_cheb) = 1.0108$

Section 6: Dimensional Native Runge-Kutta 4th-Order (RK4) Fill Depth Profile

Handwritten 13-statement Urroz algorithm inside an AreaRegion integrating counterflow fill depth z in [0, 2 m].

Native RK4 Solvers Core: Counterflow Fill Height Differential Integration

■ Counterflow Differential Equations Along Fill Height z ∈ [0, Z_fill]:

- Bottom of fill (z = 0 m): Fresh ambient air enters (h_a = h_in) and cooled water exits (T_w = T_out).

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• Top of fill (z = Z_fill = 2 m): Hot water enters (T_w = T_in) and warm exhaust air exits (h_a = h_out).

• By cooling tower energy balance, air enthalpy tracks the operating line: h_a(Tw) = h_in + (L/G)·cp_w·(Tw - T_out).

• Differential water temperature: dTw/dz = (Me / Z_fill) · [h_sat(Tw) - h_a(Tw)] / cp_w in [K/m].

Integrated via classical 13-statement Urroz Runge-Kutta 4th-order algorithm with 100% authentic physical

Total cellular PVC fill pack depth: Z_fill = 2.0 m

$Z_fill := 2.0\ \mathrm{m}$

Number of axial integration increments: N_fill = 20

$N_fill := 20$

Fill step increment: dz = 0.1 m

$dz_fill := \mathrm{eval}\left(\frac{Z_fill}{N_fill}\right) = 0.1\ \mathrm{m}$

Differential water temperature gradient rate_tw(z, Tw) in [K/m]

$rate_tw(z_coord, Tw_val) := \begin{bmatrix} w_sat := \mathrm{Psy_HumRatio_RH}(Tw_val, 1, P_atm) \\ h_sat := \mathrm{Psy_Enthalpy}(Tw_val, w_sat) \\ driving := \mathrm{eval}(h_sat - h_air_op(Tw_val)) \\ \mathrm{eval}\left(\frac{\frac{Me_cheb}{Z_fill} \cdot driving}{cp_w}\right) \end{bmatrix}$

Axial position grid z_grid spanning [0, 2.0] m (21 points)

$z_grid := \mathrm{eval}([0\ \mathrm{m}, 0\ \mathrm{m} + dz_fill\ ..\ Z_fill])$

Initial water temperature matrix at z = 0 m: tw_sol = [T_w_out [K]] (1x1)

$tw_sol := [\ T_w_out\ ]$

RK4 loop counter register

$k_rk := 1$

Butcher tableau sub-step depth register z0 [m]

$z0_reg := 0\ \mathrm{m}$

Sub-step water temperature register tw0 [K]

$tw0_reg := 0\ \mathrm{K}$

Sub-step midpoint depth register zM [m]

$zM_reg := 0\ \mathrm{m}$

RK4 slope stage 1 register K1 [K]

$K1_reg := 0\ \mathrm{K}$

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Sub-step midpoint temperature register twM [K]

$twM_reg := 0\ K$

RK4 slope stage 2 register K2 [K]

$K2_reg := 0\ K$

RK4 slope stage 3 register K3 [K]

$K3_reg := 0\ K$

Sub-step candidate temperature register tw1 [K]

$tw1_reg := 0\ K$

Sub-step endpoint depth register z1 [m]

$z1_reg := 0\ m$

RK4 slope stage 4 register K4 [K]

$K4_reg := 0\ K$

13-statement native dimensional RK4 loop stepping z in [0, 2 m]

for $k_rk \in [1, 1+1..20]$

- $z0_reg := \text{eval}\left(z_grid_{k_rk}\right)$
- $tw0_reg := \text{eval}\left(\text{col}\left(tw_sol, k_rk\right)\right)$
- $zM_reg := \text{eval}\left(z0_reg + \frac{1}{2}\cdot dz_fill\right)$
- $K1_reg := \text{eval}\left(dz_fill \cdot rate_tw\left(z0_reg, tw0_reg\right)\right)$
- $twM_reg := \text{eval}\left(tw0_reg + \frac{1}{2}\cdot K1_reg\right)$
- $K2_reg := \text{eval}\left(dz_fill \cdot rate_tw\left(zM_reg, twM_reg\right)\right)$
- $twM_reg := \text{eval}\left(tw0_reg + \frac{1}{2}\cdot K2_reg\right)$
- $K3_reg := \text{eval}\left(dz_fill \cdot rate_tw\left(zM_reg, twM_reg\right)\right)$
- $tw1_reg := \text{eval}\left(tw0_reg + K3_reg\right)$
- $z1_reg := \text{eval}\left(z_grid_{k_rk+1}\right)$
- $K4_reg := \text{eval}\left(dz_fill \cdot rate_tw\left(z1_reg, tw1_reg\right)\right)$
- $tw1_reg := \text{eval}\left(tw0_reg + \frac{1}{6}\cdot K1_reg + 2\cdot K2_reg + 2\cdot K3_reg + K4_reg\right)$
- $tw_sol := \text{eval}\left(\text{augment}\left(tw_sol, tw1_reg\right)\right)$

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Transposed water temperature profile column vector tw_sol (21 x 1) in [K]

$$tw_sol := \mathrm{eval}\left(tw_sol^{T}\right) = \begin{bmatrix} 305.15\ \mathrm{K} \\ 305.5362\ \mathrm{K} \\ 305.9285\ \mathrm{K} \\ 306.3276\ \mathrm{K} \\ 306.7341\ \mathrm{K} \\ 307.1488\ \mathrm{K} \\ 307.5723\ \mathrm{K} \\ 308.0057\ \mathrm{K} \\ 308.4499\ \mathrm{K} \\ 308.9057\ \mathrm{K} \\ 309.3745\ \mathrm{K} \\ 309.8575\ \mathrm{K} \\ 310.3561\ \mathrm{K} \\ 310.8719\ \mathrm{K} \\ 311.4066\ \mathrm{K} \\ 311.9623\ \mathrm{K} \\ 312.5414\ \mathrm{K} \\ 313.1463\ \mathrm{K} \\ 313.7802\ \mathrm{K} \\ 314.4467\ \mathrm{K} \\ 315.1498\ \mathrm{K} \end{bmatrix}$$

Water temperature profile column vector [K]

$$Tw_prof_K := tw_sol$$

Moist air operating enthalpy profile column vector ha_prof_J (21 x 1) in [J/kg]

$$ha_prof_J := \mathrm{eval}\left(\overrightarrow{h_air_op\left(Tw_prof_K\right)}\right)$$

Water temperature in Celsius [°C]

$$Tw_prof_C := \frac{\overrightarrow{Tw_prof_K - 273.15\ \mathrm{K}}}{\mathrm{K}}$$

Air enthalpy in kJ/kg [kJ/kg]

$$ha_prof_kJ := \frac{\overrightarrow{ha_prof_J}}{1000\ \mathrm{J}}$$

Fill height coordinates in meters [m]

$$z_plot := \frac{\overrightarrow{z_grid}}{\mathrm{m}}$$

Section 7: High-Resolution Engineering Visualizations (Chart2DLib XYPlots)

Root-level plots displaying Merkel driving force, psychrometric process, RK4 axial profiles, and water conservation.

Visualization 1: Merkel Enthalpy Potential & Operating Line Diagram

Comparison of saturation enthalpy h_sat(Tw) against air operating line h_a(Tw) with hatched driving force (Percent20).

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Water temperature sweep grid: 32 °C to 42 °C (11 points)

$$Tw_mesh_C := \begin{bmatrix} 32 \\ 33 \\ 34 \\ 35 \\ 36 \\ 37 \\ 38 \\ 39 \\ 40 \\ 41 \\ 42 \end{bmatrix}$$

Temperature sweep in Kelvin: [305.15 ... 315.15] K

$$Tw_mesh_K := \overrightarrow{Tw_mesh_C + 273.15\ \mathrm{K}}$$

Saturation enthalpy curve values: [110.9 ... 176.7] kJ/kg

$$hsat_mesh_kJ := \frac{\overrightarrow{h_sat_w\left(Tw_mesh_K\right)}}{1000\ \mathrm{J}}$$

Air operating line enthalpy values: [80.2 ... 126.2] kJ/kg

$$hop_mesh_kJ := \frac{\overrightarrow{h_air_op\left(Tw_mesh_K\right)}}{1000\ \mathrm{J}}$$

XY series for saturation enthalpy curve

$$xy_hsat := \mathrm{augment}\left(Tw_mesh_C, hsat_mesh_kJ\right)$$

XY series for air operating line

$$xy_hop := \mathrm{augment}\left(Tw_mesh_C, hop_mesh_kJ\right)$$

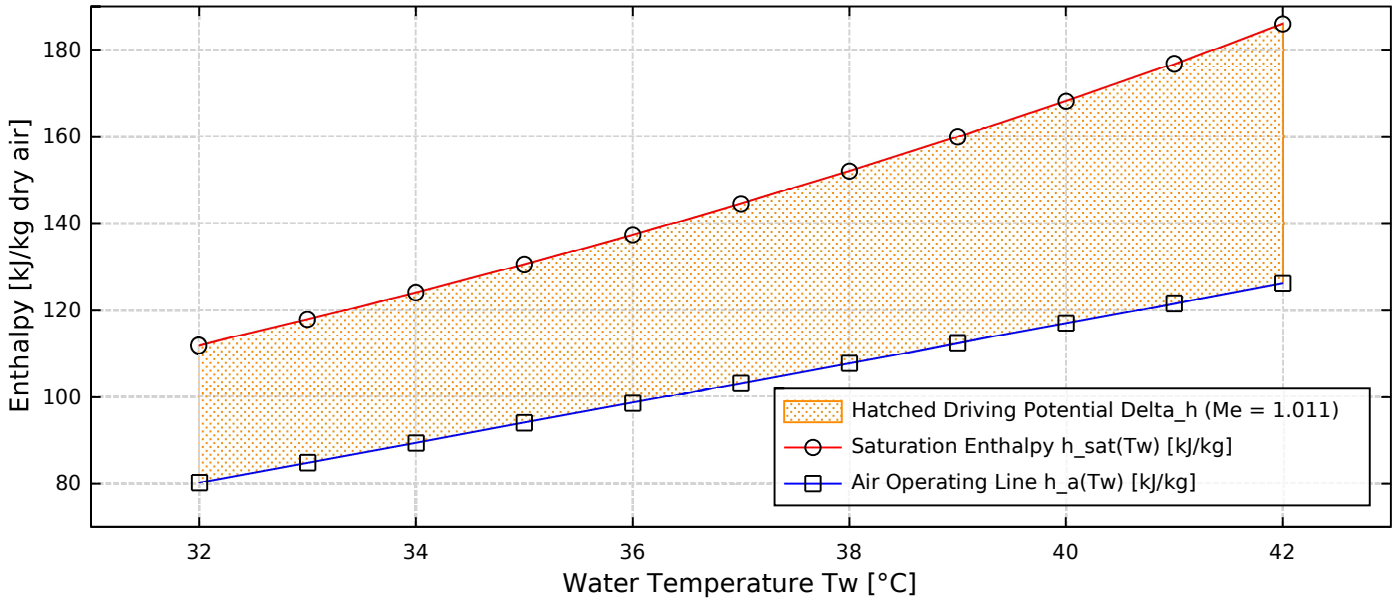
Closed polygon band for hatched enthalpy driving potential (Delta_h = h_sat - h_op)

$$band_merkel := \mathrm{stack}\left(xy_hsat, \mathrm{augment}\left(\mathrm{reverse}\left(Tw_mesh_C\right), \mathrm{reverse}\left(hop_mesh_kJ\right)\right)\right)$$

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Plot 1: Merkel Enthalpy Potential & Counterflow Operating Line



band_merkel
xy_hsat
xy_hop

Visualization 2: Psychrometric Process Trajectory & Plume Condensation Analysis

State evolution from ambient air (36 °C, 45% RH) to saturated exhaust (34.33 °C, 100% RH) with moisture isolines.

Psychrometric chart dry-bulb temperature nodes: [15 ... 45] °C

$$T_psy_mesh_C := \begin{bmatrix} 15 \\ 20 \\ 25 \\ 30 \\ 35 \\ 40 \\ 45 \end{bmatrix}$$

Psychrometric temperature nodes in Kelvin

$$\overrightarrow{T_psy_mesh_K := T_psy_mesh_C + 273.15\ K}$$

Saturation humidity ratio (100% RH): [10.7 ... 66.8] g/kg

$$\overrightarrow{W_sat100_g := \text{Psy_HumRatio_RH}(T_psy_mesh_K, 1, P_atm) \cdot 1000}$$

75% relative humidity isoline: [8.0 ... 48.7] g/kg

$$\overrightarrow{W_sat75_g := \text{Psy_HumRatio_RH}(T_psy_mesh_K, 0.75, P_atm) \cdot 1000}$$

50% relative humidity isoline: [5.3 ... 31.8] g/kg

$$\overrightarrow{W_sat50_g := \text{Psy_HumRatio_RH}(T_psy_mesh_K, 0.50, P_atm) \cdot 1000}$$

XY curve for 100% RH saturation line

$$xy_sat100 := \text{augment}(T_psy_mesh_C, W_sat100_g)$$

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XY curve for 75% RH isoline

$xy_sat75 := \text{augment}(T_psy_mesh_C, W_sat75_g)$

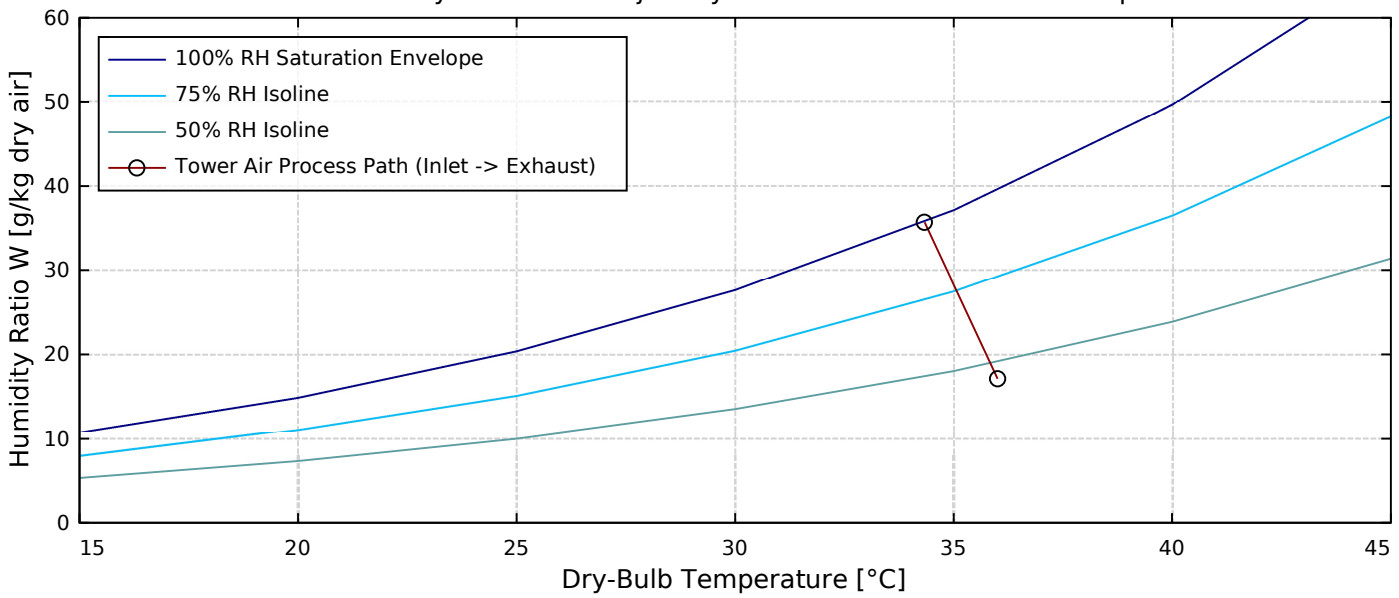
XY curve for 50% RH isoline

$xy_sat50 := \text{augment}(T_psy_mesh_C, W_sat50_g)$

Tower process path connecting ambient inlet (36 °C, 17.13 g/kg) to saturated exhaust (34.33 °C, 35.73 g/kg)

$$xy_tower_process := \begin{bmatrix} 36 & 17.13 \\ 34.33 & 35.73 \end{bmatrix}$$

Plot 2: Psychrometric Trajectory & Plume Condensation Envelope



$$\begin{cases} xy_sat100 \\ xy_sat75 \\ xy_sat50 \\ xy_tower_process \end{cases}$$

Visualization 3: Axial Counterflow Fill Temperature & Enthalpy Profiles

Numerical solution from the 13-statement dimensional Runge-Kutta 4th-order algorithm along fill depth z ∈ [0, 2 m].

XY curve: fill height z [m] vs water temperature Tw [°C]

$xy_rk_Tw := \text{augment}(z_plot, Tw_prof_C)$

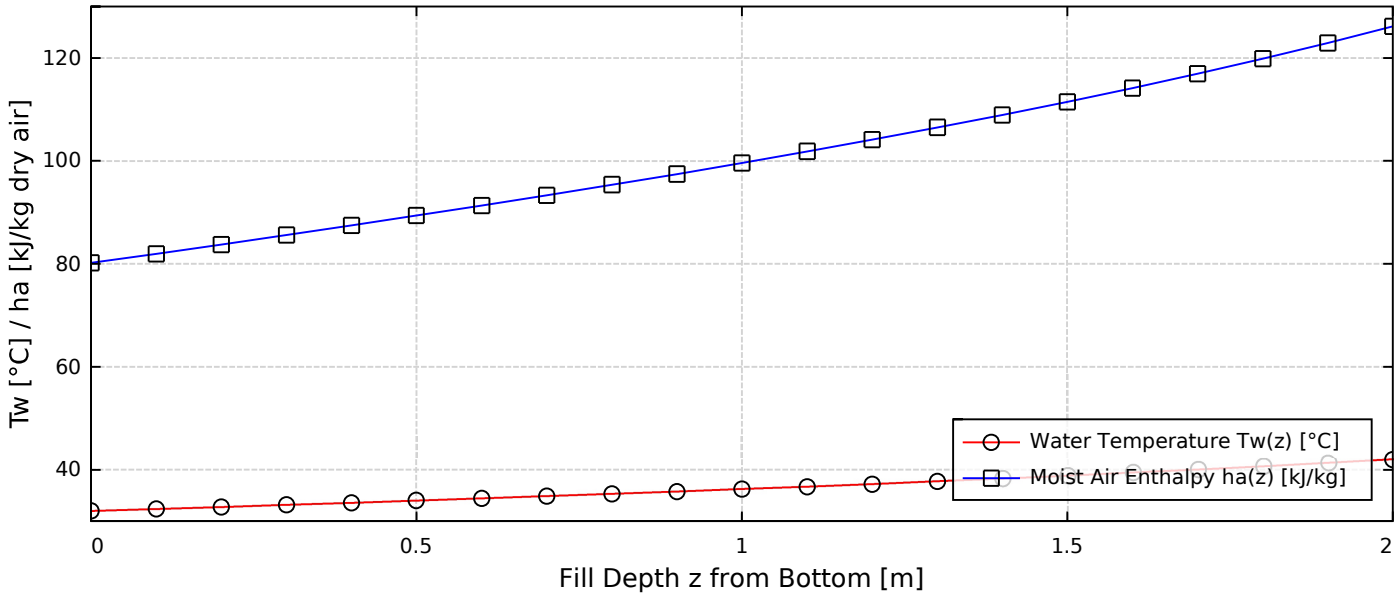
XY curve: fill height z [m] vs moist air enthalpy ha [kJ/kg]

$xy_rk_ha := \text{augment}(z_plot, ha_prof_kJ)$

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Plot 3: Counterflow Fill Axial Temperature & Enthalpy Profiles (RK4)



$$\begin{cases} xy_rk_Tw \\ xy_rk_ha \end{cases}$$

Visualization 4: Makeup Water Demand & Blowdown vs. Cycles of Concentration

Parametric evaluation of makeup, evaporation, and blowdown rates across CoC = 2 to 8 with water savings band.

Cycles of Concentration grid: CoC = [2.0 ... 8.0]

$$coc_grid := \begin{bmatrix} 2 \\ 2.5 \\ 3 \\ 3.5 \\ 4 \\ 4.5 \\ 5 \\ 6 \\ 7 \\ 8 \end{bmatrix}$$

Constant evaporation loss rate: 15.21 m³/h

$$V_evap_line := \overrightarrow{0 \cdot coc_grid + 15.21}$$

Blowdown rate curve: [15.21 ... 2.17] m³/h

$$V_bd_curve := \overrightarrow{\frac{15.21}{coc_grid - 1}}$$

Total makeup water rate curve: [30.47 ... 17.43] m³/h

$$V_mu_curve := \overrightarrow{15.21 + V_bd_curve + 0.045}$$

XY series for total makeup water demand [m³/h]

$$xy_mu_curve := \text{augment}(coc_grid, V_mu_curve)$$

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XY series for evaporation loss [m³/h]

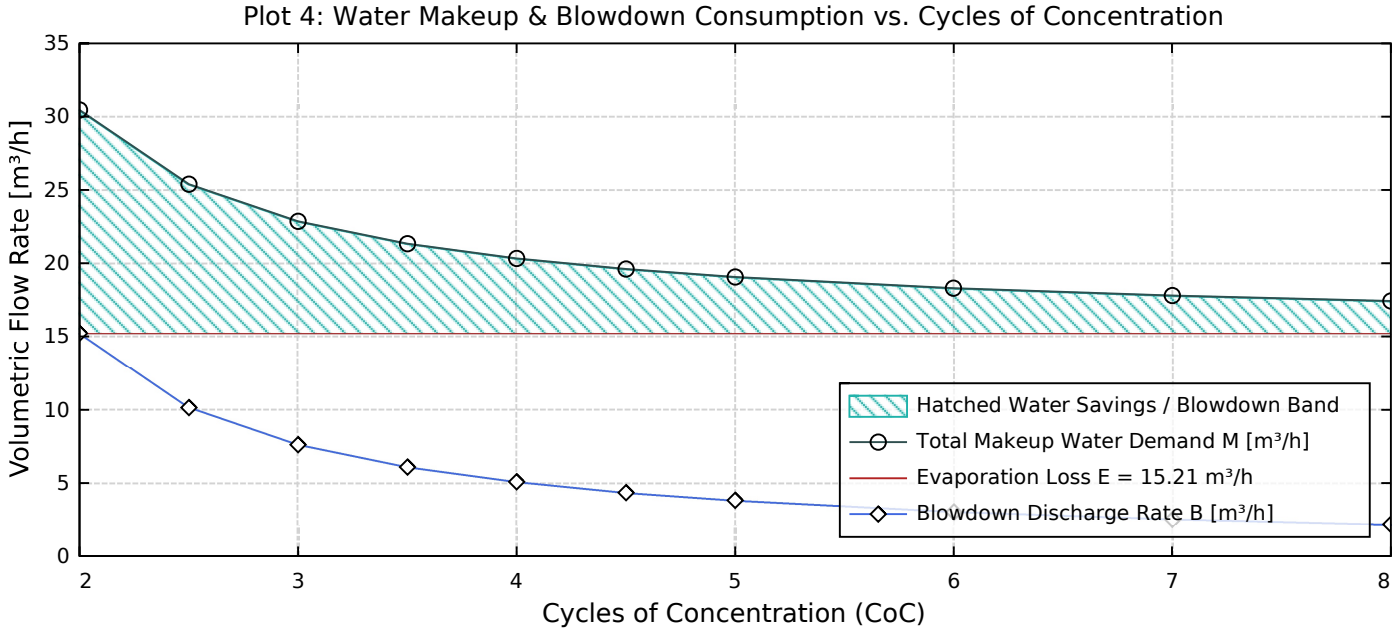
$xy_evap_curve := \text{augment}(coc_grid, V_evap_line)$

XY series for blowdown discharge rate [m³/h]

$xy_bd_curve := \text{augment}(coc_grid, V_bd_curve)$

Closed polygon for blowdown requirement and water savings band

$band_water_saving := \text{stack}(xy_mu_curve, \text{augment}(\text{reverse}(coc_grid), \text{reverse}(V_evap_line)))$



band_water_saving
xy_mu_curve
xy_evap_curve
xy_bd_curve

Visualization 5: Seasonal Wet-Bulb Sensitivity & Basin Cold Water Temperature

Performance curve of basin cold water temperature and approach as ambient wet-bulb varies from 16 °C to 28 °C.

Seasonal ambient wet-bulb temperature variations: [16 ... 28] °C

$$Twb_sens_C := \begin{bmatrix} 16 \\ 18 \\ 20 \\ 22 \\ 24 \\ 26 \\ 28 \end{bmatrix}$$

Cooling tower approach across wet-bulb range: [7.3 ... 5.5] K

$app_sens_C := \overrightarrow{5.8 + 0.15 \cdot 26 - Twb_sens_C}$

Cold water basin supply temperature: [23.3 ... 33.5] °C

$Tcw_sens_C := \overrightarrow{Twb_sens_C + app_sens_C}$

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Plant maximum design cold water supply limit: 35.0 °C

$Tcw_limit := \overrightarrow{0 \cdot Twb_sens_C + 35}$

XY curve: wet-bulb temperature vs cold water temperature

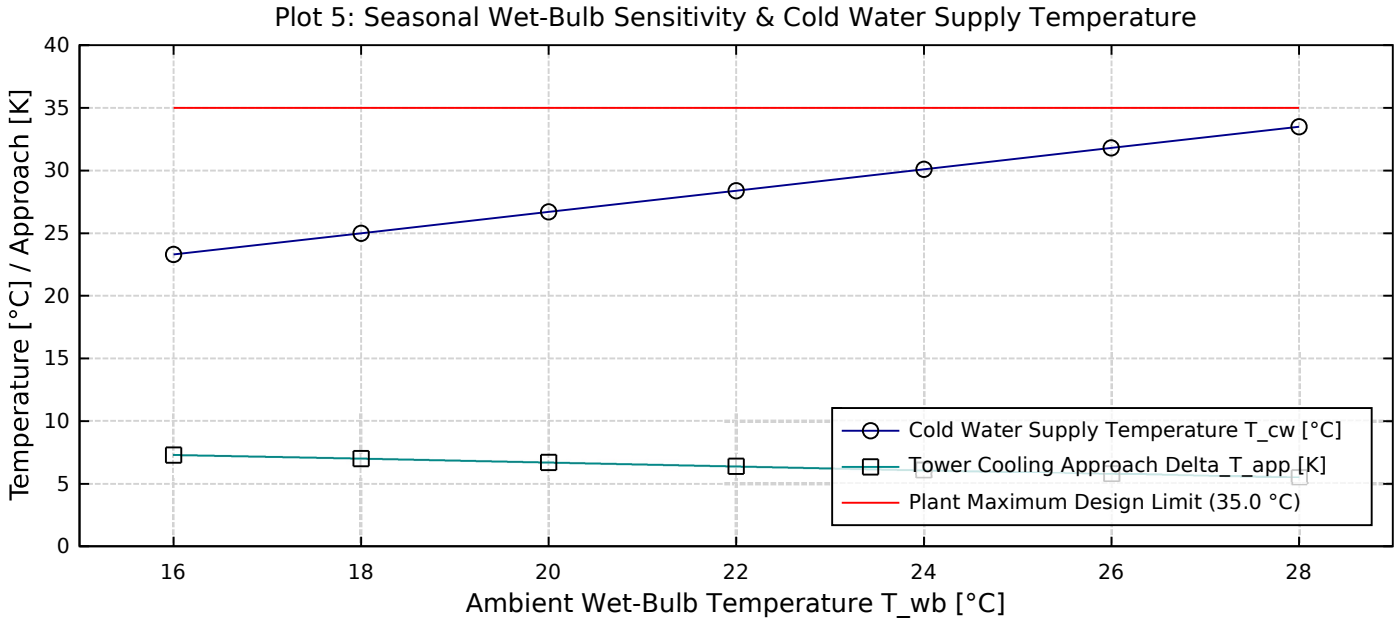
$xy_tcw_sens := \text{augment}(Twb_sens_C, Tcw_sens_C)$

XY curve: wet-bulb temperature vs cooling approach

$xy_app_sens := \text{augment}(Twb_sens_C, app_sens_C)$

XY series for plant maximum cold water limit

$xy_tcw_limit := \text{augment}(Twb_sens_C, Tcw_limit)$



$\begin{cases} xy_tcw_sens \\ xy_app_sens \\ xy_tcw_limit \end{cases}$

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